NATHAZ AI LAB / TECHNICAL REPORT · EXPLORATORY TRIAL

R23.2d Wind warnings over complex terrain

Group AI Weather Models

Status Closed (negative result)

Period 2026-10-14 15:03 — 2026-10-24 21:42

Stage conclusion

Project report

E Wind warnings in complex terrain

Objective

Assess whether the pipeline "AI mesoscale/global forecast → terrain microscale downscaling → strong-wind warnings for bridge sites / transmission lines / mountainous terrain" is worth pursuing. Deliverables: a table of prior work, gaps, and open-source code availability; a one-sentence conclusion (worth pursuing / shelved); and a minimal pilot candidate with pre-specified criteria (stations, forecast sources, downscaling methods). The PI (10-14): this must be done separately from urban wind fields, because the scales and research communities differ.

Current status

On hold (shelved 10-14 21:05). Pilot v2 failed all four frozen criteria: pooled strong-wind MAE was 6.38 m/s for bilinear interpolation vs. 6.70 m/s for WindNinja (−5.1%, winning at 3/12 stations); post-correction MAE was 3.25 vs. 3.50, showing no incremental gain; all-timesteps MAE was 3.04 vs. 3.09. The 0.25° forecasts systematically underestimated strong winds (−6.4 m/s, worst at ridge stations where station elevations sit 500–830 m above model terrain); single-station 4-sector MOS halved the error, whereas WindNinja had virtually no effect. Total compute used was ~0.42 CU. Restart conditions: pre-register before testing learning-based downscaling (FuXi-CFD/DEVINE); switch to site-specific correction + gusts if bridge-site / transmission-line data become available; connect LBM terrain priors only after learning-based downscaling demonstrates incremental value; re-run Deep Research once quota recovers.

Hierarchy and boundaries

Tier-2 node: do not open tier-3 sub-projects; use tasks for further subdivision. Task delegation is strictly limited to "this node → bot → tool," and task briefs must not allow tool bots to delegate further. Colab has no per-experiment limit (updated policy as of 10-14 18:3x; previous ≤20/≤10 CU limits are void), though any single experiment expected to exceed 100 CU must pause and report to Node A first; sessions must be stopped immediately after use, and only one executor may be dispatched per round. Single supercomputer jobs are capped at ≤50 SU (request through ComputeOps, register the job, and run pre-check and submission as two separate commands). Observations must use only public and redistributable data. Forecast sources and data retrieval must reuse A1 scripts (AIWP GraphCast/Pangu/FourCastNet v2 including 100 m u/v, GFS) without rewriting.

Prior work and gaps (Grok search + lead verification + review revisions; ✔ = personally verified by the lead; Deep Research not yet run)

Method family	Representative work	Input	Output	Gust / transient	Uncertainty	Observational validation and strong-wind performance	Open source and license
AI forecast + correction + terrain downscaling (typhoon)	ACDF, Wu et al. 2026, arXiv 2603.12828 ✓abstract	AI global forecast (benchmarked against Pangu)	500 m, Zhejiang	No	—	Leave-one-out validation across 11 typhoons; station MAE reduced by 38.8% vs. Pangu; flagged transmission lines that subsequently failed during Hagupit	Code not found
AI forecast → WRF dynamical downscaling	Brewer, Fovell & Capps 2025 *WAF* 40:187–206 ✓DOI	GraphCast 0.25°	Kilometer-scale WRF	Partial	—	Downslope windstorms such as the Marshall Fire; regional maximum sustained wind / threshold scores (full text read by reviewer A)	WRF open-source
Generative / diffusion priors	KiloGen, Cai et al. 2026, arXiv 2607.21460 ✓	25 km operational ECMWF forecast	Kilometer-scale, Shanxi	No	Posterior sampling	RMSE for winds >20 m/s reduced by ~10%; outperformed 0.25° ECMWF across all 13 strong-wind events ✓	✓ KiloGen on GitHub, MIT, no weights

Method family	Representative work	Input	Output	Gust / transient	Uncertainty	Observational validation and strong-wind performance	Open source and license
Deep learning downscaling (Alps)	DEVINE: Le Toumelin et al. 2023 *AIES*, 2024 *NPG* 31:75–97 ✓DOI; Wind-Topo (Dujardin 2022 QJRMS)	AROME / COSMO-1	30–50 m	No	None	60+ independent stations in the Alps; the 2024 paper combines statistical bias correction with terrain downscaling	DEVINE code on GitHub (not updated since 2022); Wind-Topo on EnviDat (unverified)
Diagnostic physics (mass-consistent)	WindNinja: Wagenbrenner 2016 ACP; Seto et al. 2025 WAF ✓	HRRR/NAM, etc.	100–500 m	No	None	Santa Ana case study: skill degrades during strong winds, lee-slope canyon negative bias amplified ✓ (single region, not generalized)	✓ firelab/windninja, public domain
CFD/RANS surrogate	FuXi-CFD, Lin et al. 2026 *Nat. Commun.* ✓	**100 m** wind, 9×9, 1 km	30 m 3D wind + TKE, 27 levels	No: steady-state, neutral	None	Independent validation at European tall towers (OPE, etc.), **driven by ERA5**, improvements in moderate-to-strong winds ✓	✓ Official Zenodo 10.5281/zenodo.18770845 (author's Hugging Face mirror), CC BY-NC 4.0
Mesoscale → LES	Liu 2011 JWEIA; HKIA WRF–LES (QXGRWK9H); Hong Kong bridge-site typhoon Lo 2024 (unverified)	NWP	10–100 m	Yes	Ensemble only	—	Computationally expensive

Gaps (narrowed after review): Precedents already exist for terrain downscaling directly driven by AI global forecasts in typhoons (ACDF) and dynamical WRF (Brewer); what has not yet been seen is **cold-season / non-typhoon complex terrain + fast downscaling (mass-consistent or CFD surrogate) + a verification period independent of model training + comparison against single-station statistical bias correction**. Gusts and engineering warning criteria remain lacking. Medium confidence (searched by Grok + 3 reviewers, non-exhaustive).

Minimal pilot candidate v2 (re-frozen after review, 10-14 15:55; no changes prior to execution)

- **Observations**: MeteoSwiss SwissMetNet open data (Source: MeteoSwiss). 8–12 stations spanning ridges, passes, valley floors, and plateaus; station list frozen prior to computing any forecast errors (including true elevation and elevation difference relative to 0.25° model terrain).
- **Period**: 2023-10-01 to 2024-03-31; Oct–Dec serves as the training period (MOS fitting, thresholds), and 2024-01 to 03 as the verification period.
- **Forecast sources**: Primary evaluation on FOUR_v200_GFS (which provides both 10 m and 100 m wind) at 24 h lead time; GraphCast, Pangu, GFS, and 48 h lead time are reported only and not used for gate evaluation. Initialized at 00/12 UTC, valid at 00/12 UTC.
- **Observation pairing**: Mean of the six 10-minute average wind measurements within ±30 minutes of the valid time; timestamp convention to be documented after verifying against MeteoSwiss metadata.
- **Methods**: (0) Bilinear interpolation to station (10 m); (1) **WindNinja (primary method)**: ~10 km × 10 km domain per station, mass-consistent solver, domain-average initialization using the station forecast wind vector from (0), SRTM 30 m, 100 m grid spacing, 10 m output; (2) FuXi-CFD (report only, non-evaluative): official Zenodo weights, taking FOUR/GFS 100 m winds interpolated onto a 9×9, 1 km grid and selecting the output level closest to anemometer height; GraphCast/Pangu lack 100 m winds and are excluded from (2); (3) MOS: per station × source × lead time, 4 wind-direction-sector linear bias correction (wind speed) fitted on the training period; (3') WindNinja+MOS: the same correction scheme applied to (1).
- **Strong-wind timesteps**: Verification-period observations ≥ the station's **training-period** observed P90; pooled across stations, with sample counts reported.
- **Primary criteria (all three must be satisfied = worth pursuing)**: 1. Pooled strong-wind MAE: (1) reduces MAE by ≥15% compared to (0), and (1) achieves lower MAE than (0) at ≥2/3 of stations 2. Pooled strong-wind MAE: (3') < (3) (terrain downscaling provides incremental gain beyond statistical bias correction) 3. Guardrail: Pooled all-timesteps MAE for (1) is no worse than (0).
- **Report only (non-evaluative)**: Other sources/lead times, FuXi-CFD, breakdown by terrain category, hit rate / false alarm ratio conditioned on forecast threshold exceedance (threshold set at training-period P99), and training cutoff dates of each source.
- **Scope of conclusion**: Speaks only to mean wind errors at the selected stations and valid times; does not imply validity for gusts or operational engineering warnings.
- **Budget**: No per-experiment cap on Colab (report to Node A first if projected >100 CU), only one executor per round, stop immediately after use.

Parameter source table

Parameter	Value	Source or "assumption"	Sources checked
FuXi-CFD input	**100 m** horizontal wind, 9×9, 1 km; terrain/roughness 300×300 (30 m)	Nat. Commun. 2026 Methods ✓, official HF README ✓ (arXiv v1 stated 10 m; journal version takes precedence)	arXiv v1, third-party repos, official HF
FuXi-CFD applicability	Steady-state, neutral stratification	arXiv v1 HTML ✓	—
WindNinja configuration	~10 km domain per station, mass-consistent, station forecast wind vector as domain-average initialization, 100 m grid	**Assumption** (pinned down per review request; official FAQ item 18 distinguishes two initialization modes)	WindNinja official FAQ and solver documentation (cited by reviewer A)
Strong-wind timestep threshold	Station training-period (Oct–Dec) P90	**Assumption** (large inter-station wind climatology differences make fixed m/s thresholds incomparable)	No literature basis; based on this line's practice (same quantile convention as Node H)
Criterion 15% / 2/3 stations	—	**Assumption**, same order of magnitude as Node H r2 criteria (this line's data)	—

Finalized decisions

- 2026-10-14 Pilot v2 failed criteria; node shelved
- 2026-10-14 Review verdict: retracted "no complete precedent," added Brewer/ACDF; pilot criteria v2
- 2026-10-14 Worth pursuing; run Swiss cold-season pilot first, criteria frozen

Trials and results

Date	Action	One-sentence result
2026-10-14	Pilot v2 (frozen criteria, ~0.30 CU)	**Failed**: strong-wind MAE bilinear 6.38 / WindNinja 6.70 / bilinear+MOS 3.25 / WindNinja+MOS 3.50; all-timesteps 3.04 / 3.09; won at 3/12 stations; 207 strong-wind cases, all 4,295 WindNinja runs succeeded

Date	Action	One-sentence result
2026-10-14	Legacy pilot (v1 criteria, ~0.12 CU)	12 stations frozen (3 each for ridge/pass/valley floor/plateau); WindNinja (±0.015° ~3 km domain) median strong-wind MAE change vs. interpolation was −2.9% (i.e., slightly worse), winning at 4/12 stations, 0/12 stations no worse than MOS; all four 0.25° forecast sources systematically underestimated ridge strong-wind timesteps by ~13–15 m/s, while MOS halved the error; FuXi-CFD official weights (Zenodo→HF, SHA256 recorded) passed sample inference, not run on Swiss stations. v1 not used for verdict; v2 governs
2026-10-14	Grok quick search + primary source check	Found no complete precedent for "AI global forecast → terrain downscaling → independent station strong-wind verification"; WindNinja degrades under strong winds; FuXi-CFD assumes neutral steady-state, weights initially seen only in third-party mirrors (not used; official weights only)
2026-10-14	Inquired with digital-twin wind tunnel team on terrain LBM	Not done previously; can take on in mid-to-late November, limited to neutral stratification, single wind direction, and stepped terrain
2026-10-14	Kickoff: database check, Grok quick search dispatched, inquired with digital-twin wind tunnel team	Only two relevant cards in the repository (KiloGen and HKIA)

Open items and next steps

- 10-15 02:00 Node A verification: the Colab CPU session for this pilot is no longer in the session list, so no further action is needed to stop it; E is shelved and will no longer use Colab.
- Currently shelved; see decision log for restart conditions. Frozen station table, observations, data retrieval, and WindNinja scripts are all preserved and ready for direct reuse.

Literature basis

- Seto et al. 2025, *Weather and Forecasting* 40:525–541 (abstract, no page numbers on web version): WindNinja skill degrades under strong winds.

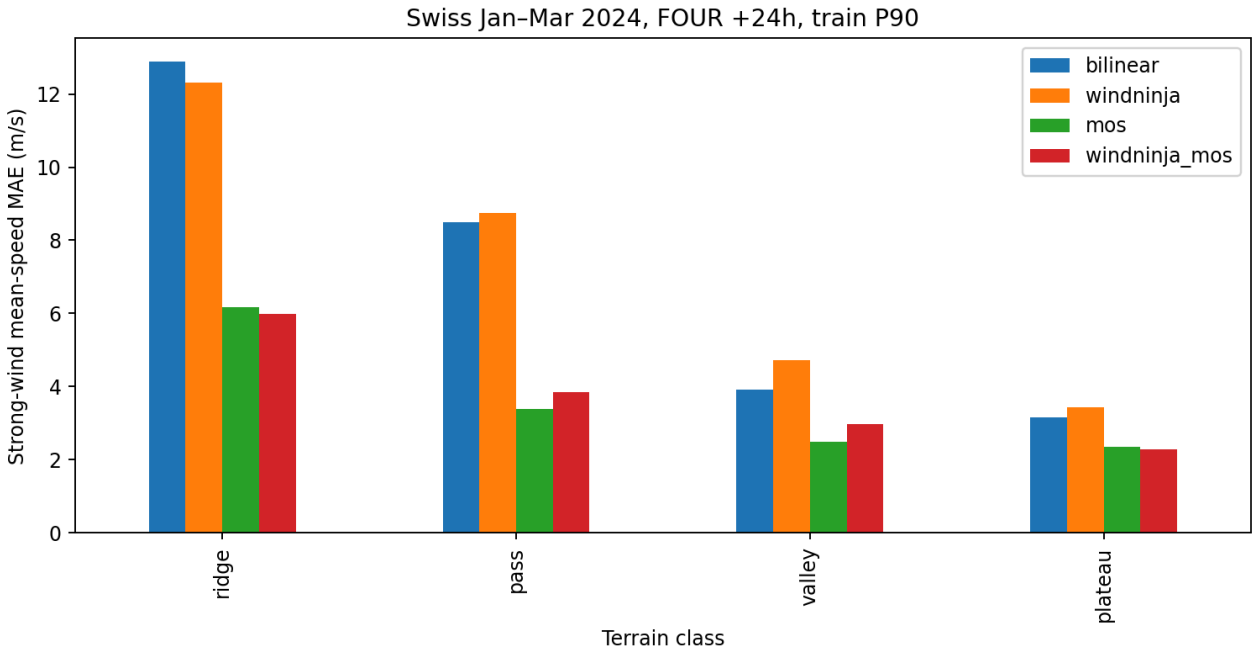
- Lin et al. arXiv 2505.12732v1 (HTML, no page numbers): FuXi-CFD input specifications and neutral steady-state limitations.
- Cai et al. arXiv 2607.21460 (abstract): KiloGen strong-wind improvement of ~10%.
- (No page numbers on card; arXiv 2607.21460v1 abstract)
- HKIA terrain wind shear WRF–LES card

Notes from the PI

Reference (forwarded by the CFD line on 10-25, read-only)

- Zhang/Yang/He 2026 JCP 557:114835 differentiable wall-modeled LES, section "Transfer assessment 1: complex-terrain wind" of the close-reading card.
- The original paper trains and tests on smooth-wall hills and bump/hump/ramp separated flows, making it a closer fit for mountain winds than for buildings; it still preserves the separation bubble on coarse grids.
- Limitations: Separation point still exhibits bias; gains vanish when flow regimes are dissimilar; no rough terrain surface or thermal stratification; code is not publicly available.
- R23.2d is currently On hold; it will be merged into the new plan prioritization as part of the R23.2 restructuring. If the downscaling layer (M5) considers CFD-type physical downscaling, contact R16.5.

Key figures



Source: MeteoSwiss; NOAA AIWP; NOAA GFS

strong mae by terrain v2